ALA-2

Topic: if-elif-else Ladder

1. What is an if-elif-else ladder?

Answer:

It is a control structure used to check multiple conditions one by one. When one condition becomes true, its block executes and the rest are skipped.

Syntax:

```
if condition1:
    statement1
elif condition2:
    statement2
else:
    statement
```

Example:

```
x = 20

if x < 10:
    print("Small")
elif x < 30:
    print("Medium")
else:
    print("Large")
```

2. Where is if-elif-else used?

Answer:

It is used in decision-making situations where multiple conditions need to be checked (like grading, menu selection, billing systems).

Syntax:

```
if condition:
    statement
elif condition:
    statement
else:
    statement
```

Example:

```
marks = 85

if marks >= 90:
    print("A")
elif marks >= 70:
    print("B")
else:
    print("C")
```

3. What is the difference between if and if-elif-else?

Answer:

- if → checks only one condition
- if-elif-else → checks multiple conditions

Syntax:

```
if condition:
    statement
```

 # vs

```
if condition:
    statement
elif condition:
    statement
```

Example:

```
x = 5

if x > 10:
    print("Greater")

# Using ladder
if x > 10:
    print("Greater")
elif x == 5:
    print("Equal to 5")
```

4. How many elif statements can we use?

Answer:

We can use multiple elif statements, there is no fixed limit.

Syntax:

```
if condition1:
    statement
elif condition2:
    statement
elif condition3:
    statement
```

Example:

```
num = 2

if num == 1:
    print("One")
elif num == 2:
    print("Two")
elif num == 3:
    print("Three")
```

5. Is else compulsory in if-elif ladder?

Answer:

No, else is optional. It is used when none of the conditions are true.

Syntax:

```
if condition:
    statement
elif condition:
    statement
```

Example:

```
x = 15
if x > 20:
    print("Greater than 20")
elif x > 10:
    print("Greater than 10")
```

6. What happens when multiple conditions are true?

Answer:

Only the first true condition executes, and the rest are ignored.

Syntax:

```
if condition1:
    statement
elif condition2:
    statement
```

Example:

```
x = 50
if x > 10:
    print("First condition")
elif x > 20:
    print("Second condition")
```

7. Can we use nested if inside if-elif?

Answer:

Yes, we can place an if inside another if or elif.

Syntax:

```
if condition:
    if condition:
        statement
elif condition:
        statement
```

Example:

```
x = 10

if x > 5:
    if x < 20:
        print("Between 5 and 20")
elif x > 20:
        print("It's bigger than 20")
```

8. What is the flow of execution in if-elif ladder?

Answer:

Conditions are checked from top to bottom. Execution stops when a true condition is found.

Syntax:

```
if condition1:
    statement
elif condition2:
    statement
else:
    statement
```

Example:

```
x = 30

if x < 10:
    print("A")
elif x < 20:
    print("B")
else:
    print("C")
```

9. Why is indentation important in if-elif-else?

Answer:

Indentation defines which statements belong to which condition block.

Syntax:

```
if condition:
    statement
```

Example:

```
x = 10

if x > 5:
    print("Correct")
    print("Block")
```

10. Can if-elif-else be replaced with other structures?

Answer:

Yes, sometimes it can be replaced with:

- Dictionary mapping
- Match-case (Python 3.10+)

Syntax:

```
match value:
    case pattern:
        statement
```

Example:

```
day = 1

match day:
    case 1:
        print("Monday")
    case 2:
        print("Tuesday")
```